TRANSPORT OF RESPIRATORY GASES

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Lecturer

DIKIOHS,DUHS.

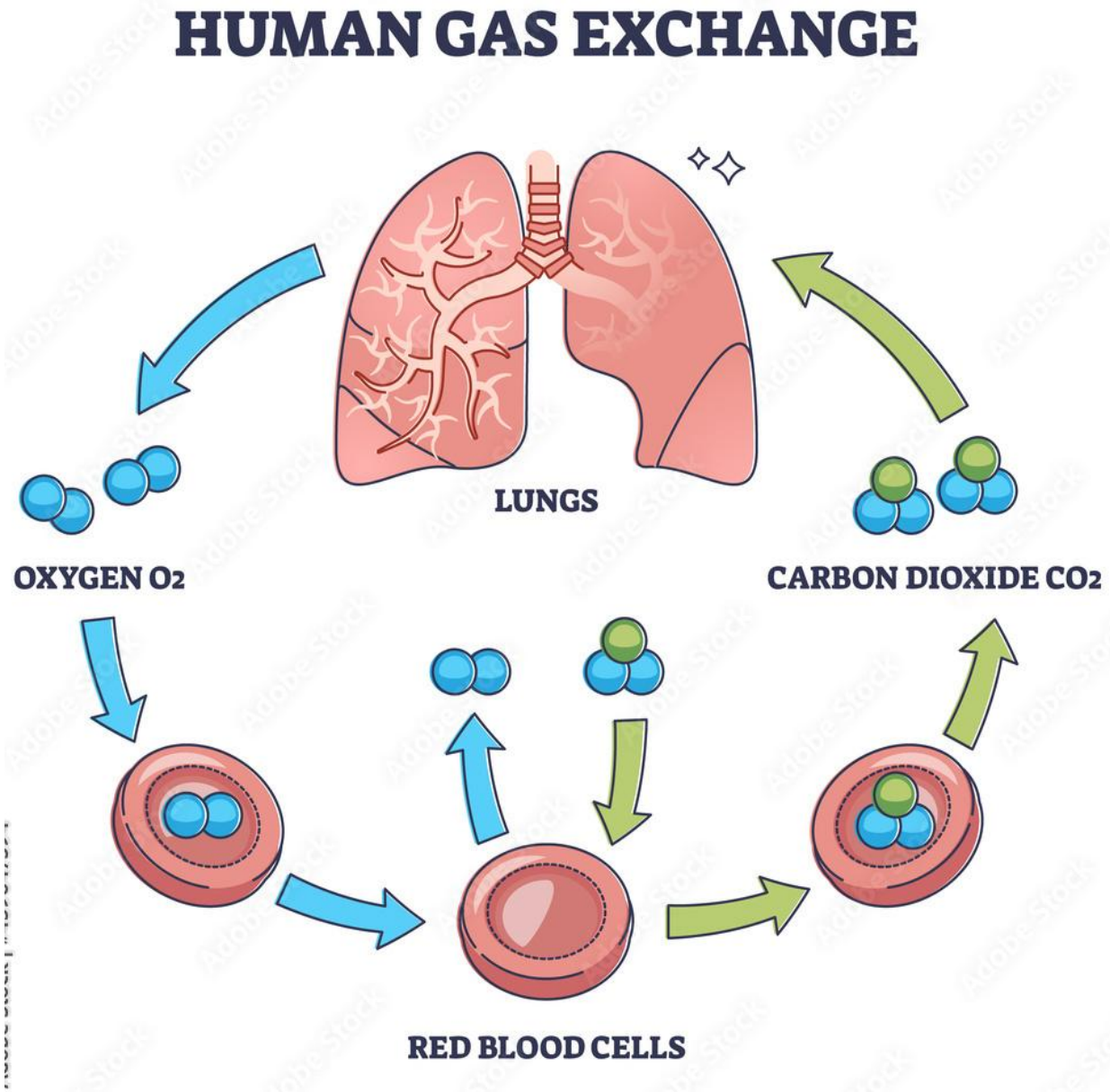
LEARNING OBJECTIVES

By the end of this lecture, students should be able to:

- ✔ Describe the transport of oxygen & carbon dioxide in the blood.
- ✔ Explain different forms of oxygen transport including oxygen bound to hemoglobin & oxygen dissolved in plasma.
- ✔ Explain oxygen-hemoglobin dissociation curve.
- ✔ Identify the factors that affect oxygen hemoglobin dissociation curve.
- ✔ Explain the mechanism of carbon dioxide transport as dissolved CO_2, bicarbonate ions & carbamino compounds.
- ✔ Explain the role of hemoglobin in the transport of oxygen & carbon dioxide.
- ✔ Define Bohr effect & Haldane effect in gas transport.

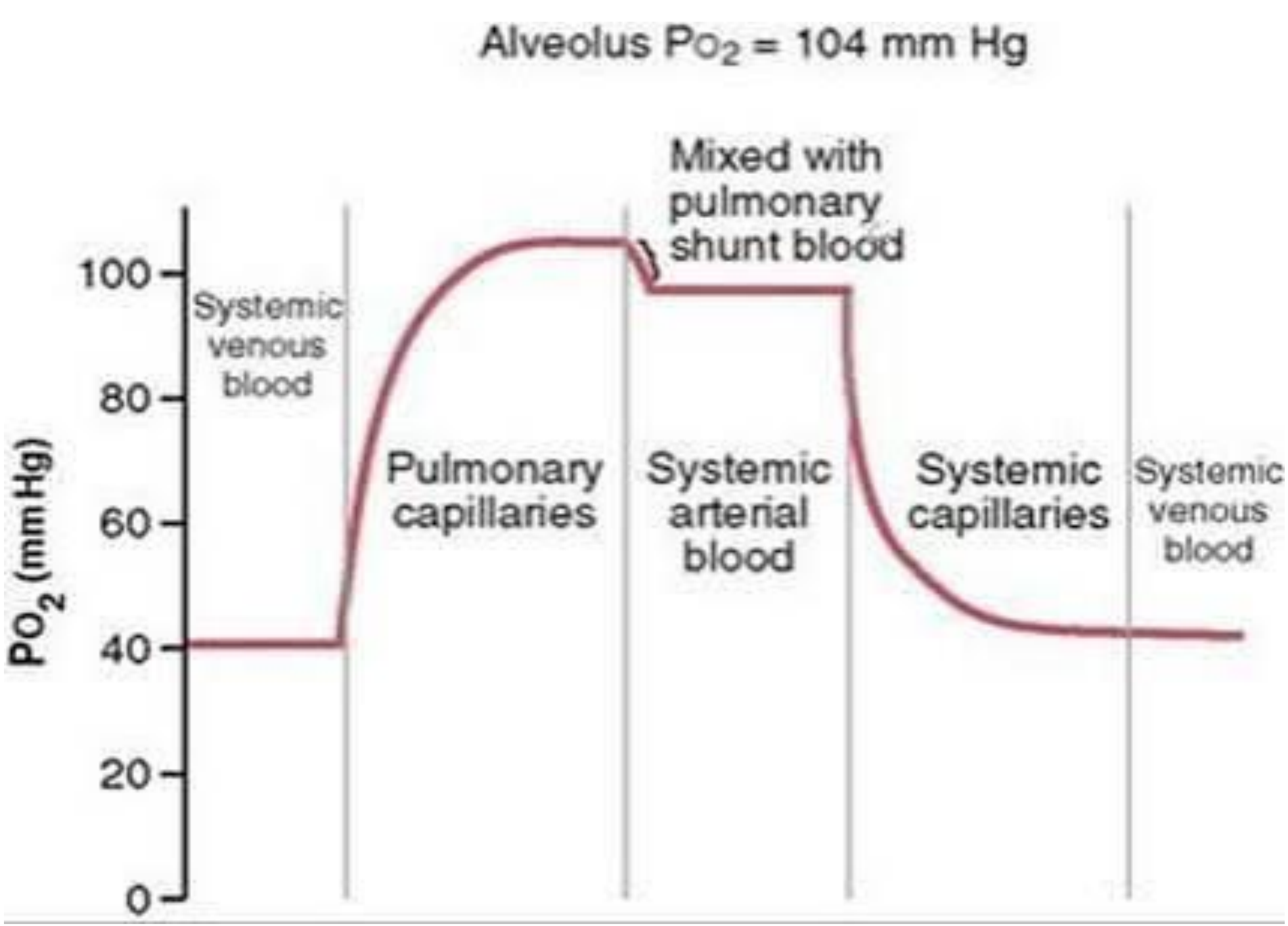
INTRODUCTION

- Blood serves to transport the respiratory gases i.e. O_2 (which is essential for the cells) & CO_2 (produced as waste product in the cells).
- O_2 is transported from alveoli of lungs to the cells while CO_2 is transported from cells to the lungs.



PARTIAL PRESSURE OF OXYGEN IN ARTERIAL BLOOD

- Gases move from one point to another by means of ***diffusion*** & cause of this movement is always a ***partial pressure difference*** between the two points.
- 98% of blood entering the right atrium from the lungs is oxygenated upto PO_2 of about ***104mm Hg*** while 2% of blood that passed through the bronchial circulation is not exposed to lung ventilation (***shunt flow***).
- PO_2 of shunt blood is approximately equals to normal systemic venous blood i.e. ***40 mm Hg***.
- Combination of shunt blood with oxygenated blood from alveolar capillaries is called ***venous admixture of blood*** & causes PO_2 of blood entering the left heart to fall (about 95mm Hg).


Alveolus P_{O_2} = 104 mm Hg
Mixed with pulmonary shunt blood
100
80
60
40
20
0
P_{O_2} (mm Hg)
Systemic venous blood
Pulmonary capillaries
Systemic arterial blood
Systemic capillaries
Systemic venous blood

DIFFUSION OF OXYGEN FROM PERIPHERAL CAPILLARIES INTO TISSUE FLUID

- Arterial blood reaching the peripheral tissues has PO_2 equals to ***95mm Hg*** whereas PO_2 in interstitial fluid surrounding the tissue cells averages only ***40 mm Hg***.
- Thus, large initial pressure difference causes O_2 to diffuse rapidly from capillary blood into the tissues & fall of capillary PO_2 almost equal to ***40 mm Hg*** (i.e. the blood entering the systemic veins).
- Oxygen is continuously being used by the cells which causes intracellular PO_2 in peripheral tissues to remain lower than PO_2 in peripheral capillaries.
- There is also considerable physical distance between the capillaries & the cells.
- Therefore, normal intracellular PO_2 ranges from as low as ***5 mm Hg*** to as high as ***40 mm Hg*** (***average 23 mm Hg***).
- As only ***1 to 3 mm*** Hg of O_2 pressure is normally needed for the chemical processes in the cell, this low intracellular PO_2 of ***23 mm Hg*** is not only more than adequate but also provides a large safety factor.

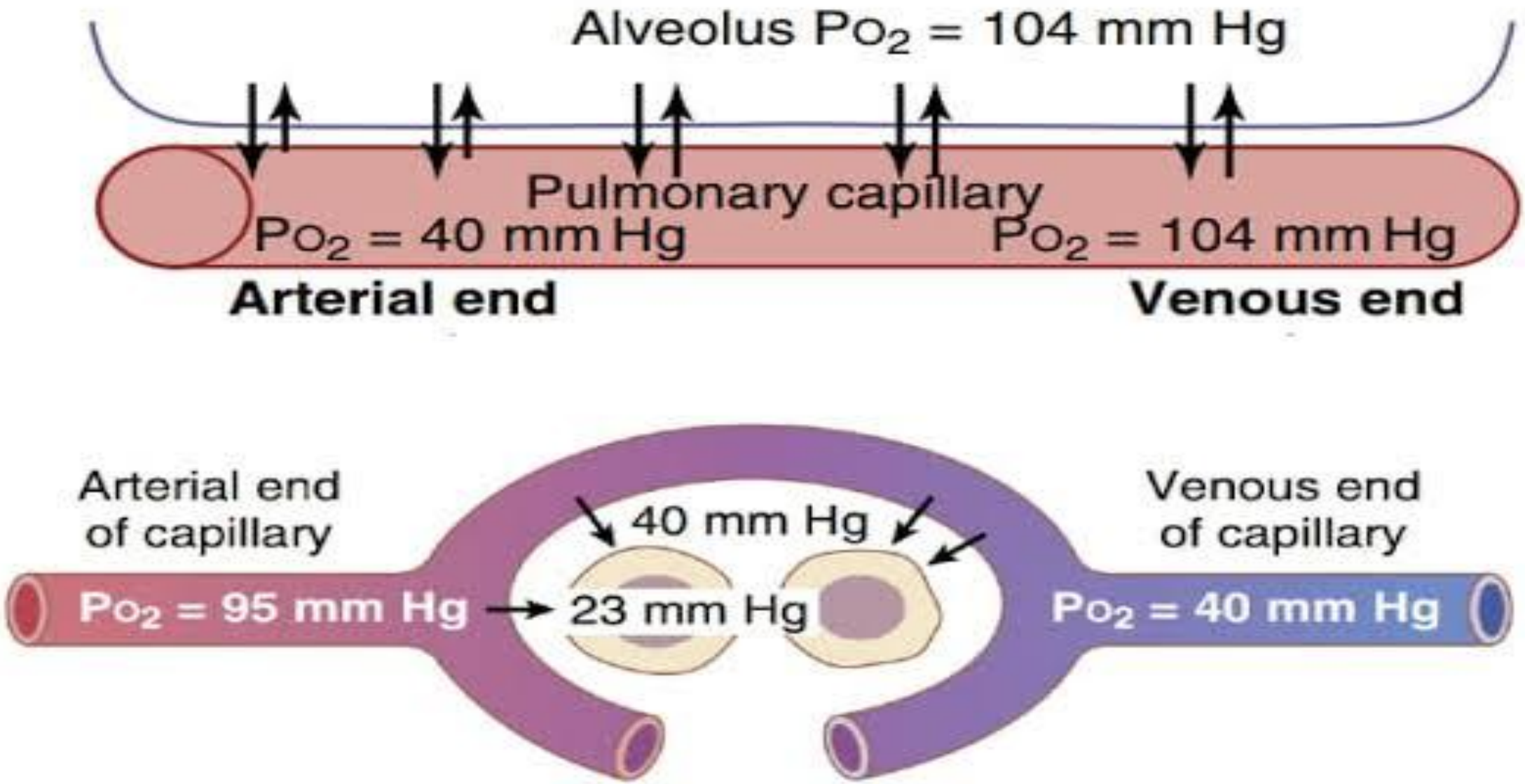


Figure 41.3 Diffusion of oxygen from a peripheral tissue capillary to the cells. (P_{O_2} in interstitial fluid = 40 mm Hg; in tissue cells, P_{CO_2} = 23 mm Hg.)

EFFECT OF BLOOD FLOW & TISSUE METABOLISM ON INTERSTITIAL FLUID PO_2

- Tissue PO_2 is affected by two factors i.e.

i. Rate of O_2 transport to the tissues from the blood

ii. Rate at which the O_2 is consumed by the tissues

- If the blood flow through a particular tissue is increased, greater amount of O_2 is transported into the tissue & the tissue PO_2 becomes correspondingly higher & if blood flow through the tissue is decreased, tissue PO_2 also decreases (point C).
- An increase in flow to 400% of normal increases PO_2 from ***40 mm Hg*** (point A) to ***66 mm Hg*** (point B).
- However, maximum limit to which PO_2 can rise even with maximum blood flow is ***95 mm Hg*** (as this the O_2 pressure in arterial blood).
- Similarly, If the cells consume more O_2 for metabolism than normal, interstitial fluid PO_2 is decreased.

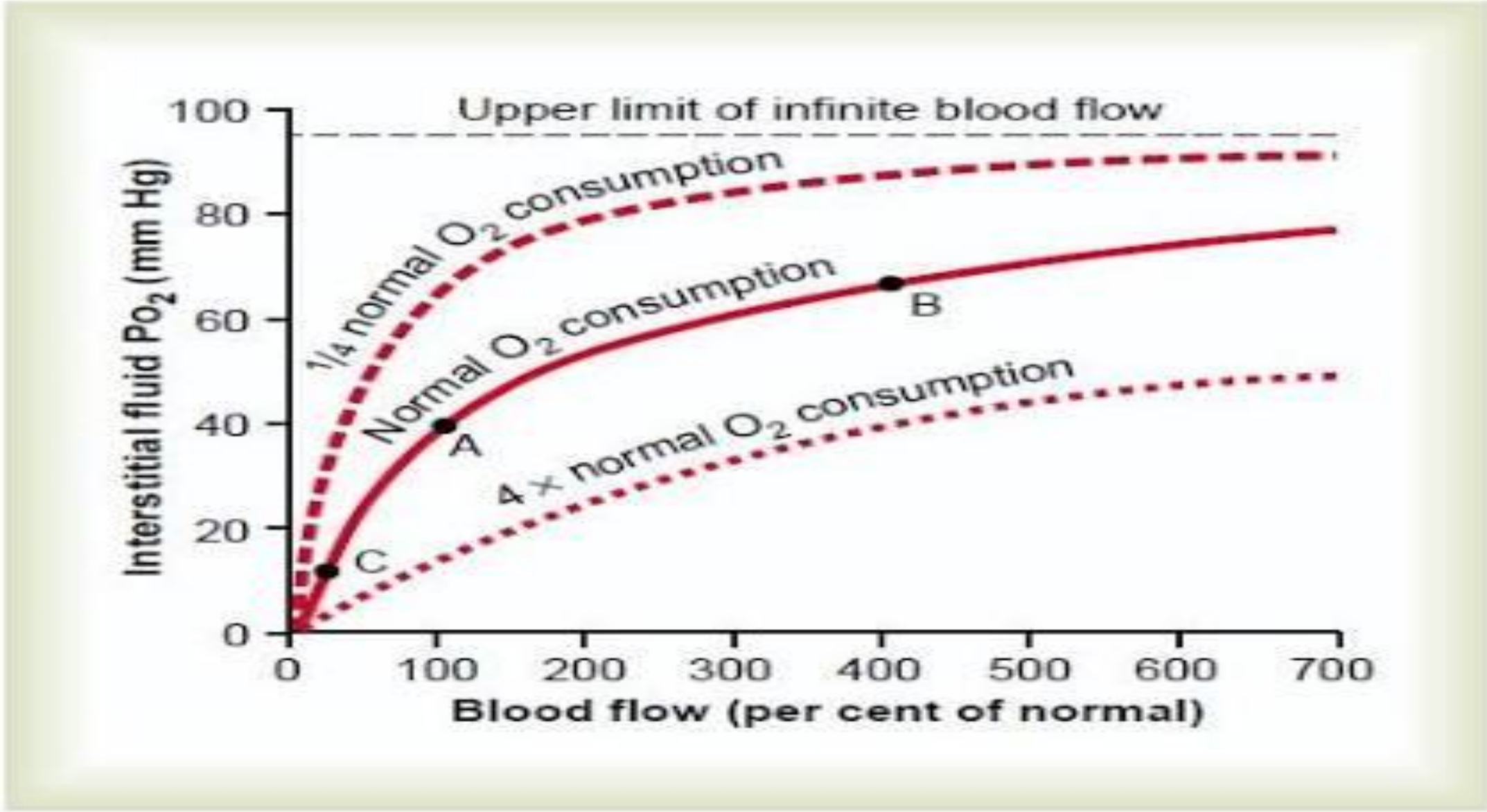


Figure 40–4

Effect of blood flow and rate of oxygen consumption on tissue Po_2.

TRANSPORT OF OXYGEN

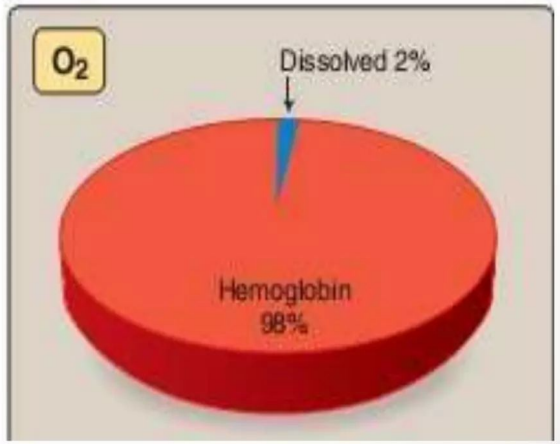
AS SIMPLE SOLUTION

- Oxygen dissolves in water of plasma & is transported in this physical form.
- However, amount of oxygen transported in this way is very negligible i.e. only ***0.3 ml/100 ml*** of plasma & constitutes only about ***3%*** of total oxygen in the blood (due to poor solubility of oxygen in water content of plasma).
- Nevertheless, transport of oxygen in this form is important especially during conditions like muscular exercise to meet the excess demand of oxygen by the tissues.

TRANSPORT OF OXYGEN :

- Oxygen is transported from alveoli to the tissue by blood in two forms:

1. **As simple physical solution (3%).**
2. **In combination with hemoglobin (97%).**



TRANSPORT OF OXYGEN IN COMBINATION WITH HEMOGLOBIN

- Oxygen combines with hemoglobin in the blood & transported as ***oxyhemoglobin***.
- Transport of oxygen in this form is particularly important as maximum amount is transported in this way.

OXYGENATION OF HEMOGLOBIN

- Oxygen combines with hemoglobin only as a physical combination (i.e. only ***oxygenation*** & not oxidation).
- The advantage of this combination is that oxygen can be readily released from hemoglobin when it is needed.
- Hemoglobin accepts oxygen readily whenever the partial pressure of oxygen in the blood is more & gives out oxygen whenever the PO2 in the blood falls.
- Oxygen combines with the ***iron in heme part*** of hemoglobin & each molecule of hemoglobin contains 4 atoms of iron.
- Each iron atom combines with one molecule of oxygen & remain in ***ferrous form***.

Hemoglobin = 4 Polypeptide Chains

Each chain contains a **heme group**



Hemoglobin = Oxygen Carrier

Each chain contains a **heme group**

Iron (Fe^{2+})
binds oxygen



1 Hemoglobin = Up to 4 O_2

OXYGEN CARRYING CAPACITY OF HEMOGLOBIN

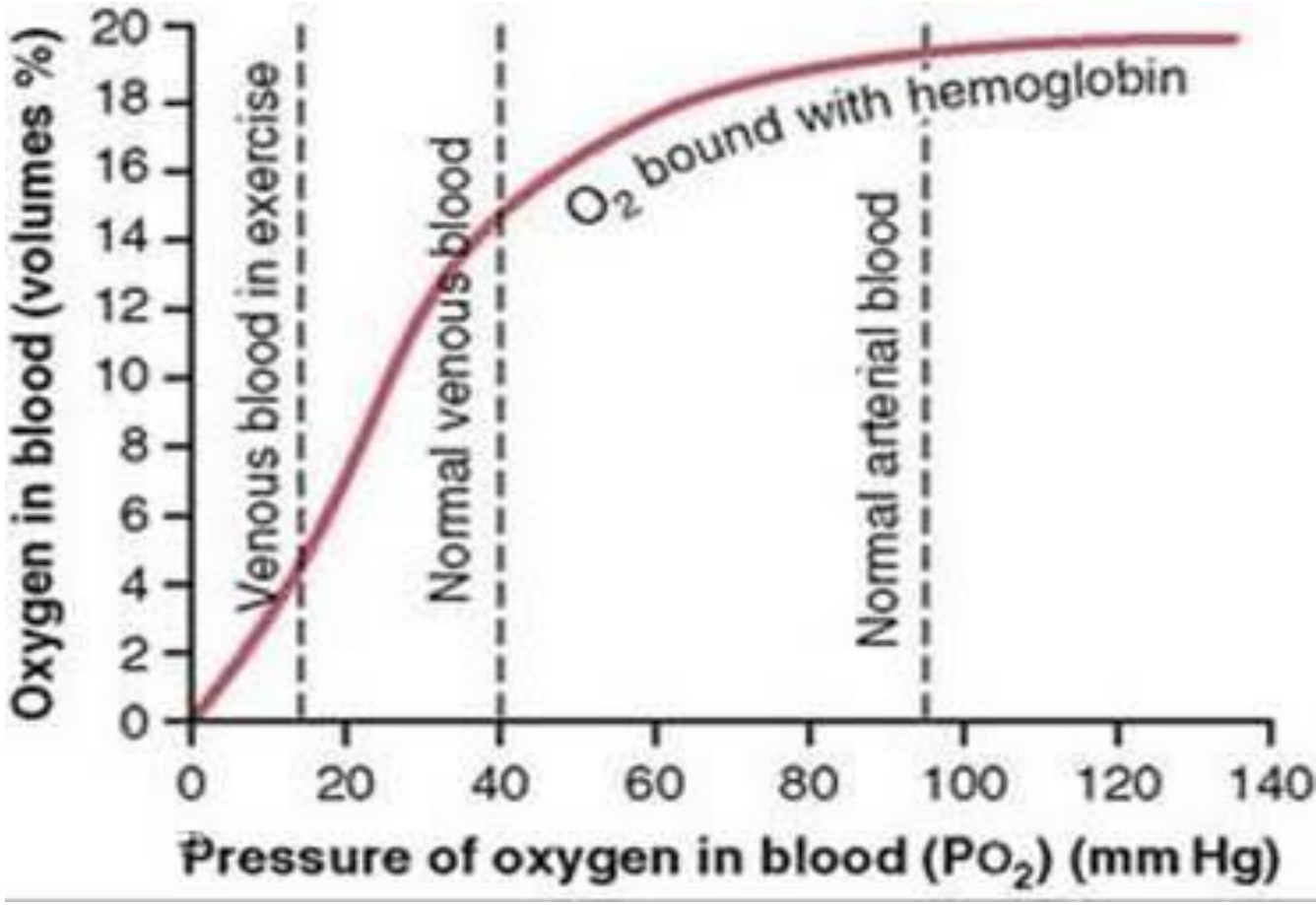
- Amount of oxygen transported by 1 gram of hemoglobin.
- It is ***1.34 ml/gm***.

OXYGEN CARRYING CAPACITY OF BLOOD

- It is the amount of oxygen transported by blood.
- Normal hemoglobin content in blood is ***15 g/100 ml***.
- Therefore 15 g of hemoglobin in 100 ml of blood can combine with a maximum of about 20 ml of oxygen (hemoglobin = 100% saturated), this is expressed as ***20 volume percent***.
- However, total quantity of oxygen bound with hemoglobin in normal systemic arterial blood is ***19.4 ml/100 ml*** of blood (hemoglobin = 97% saturated).

HEAVY EXERCISE INCREASES OXYGEN TRANSPORT

- During strenuous exercise, muscle cells consume oxygen at a rapid rate which in extreme cases cause interstitial fluid PO_2 to fall from 40mm Hg to 15 mm Hg.
- At this PO_2, only 4.4 ml of oxygen remains bound to hemoglobin in each 100 ml of blood.
- Thus ***15 ml (19.4 – 4.4 ml)*** of oxygen is delivered to the tissues by each 100 ml of blood.



UTILIZATION COEFFICIENT

- It is the % of blood that gives up its oxygen while passing through tissue capillaries.
- Normal value is ***25%*** i.e. 25% of oxygenated hemoglobin gives its oxygen to the tissues.
- However, during intense exercise, utilization coefficient in the body can increase to 75% to 85%.
- Whereas, in tissue areas where blood flow is extremely slow or metabolic rate is very high, utilization coefficient may rise upto 100% i.e. essentially all the oxygen is given to tissues.

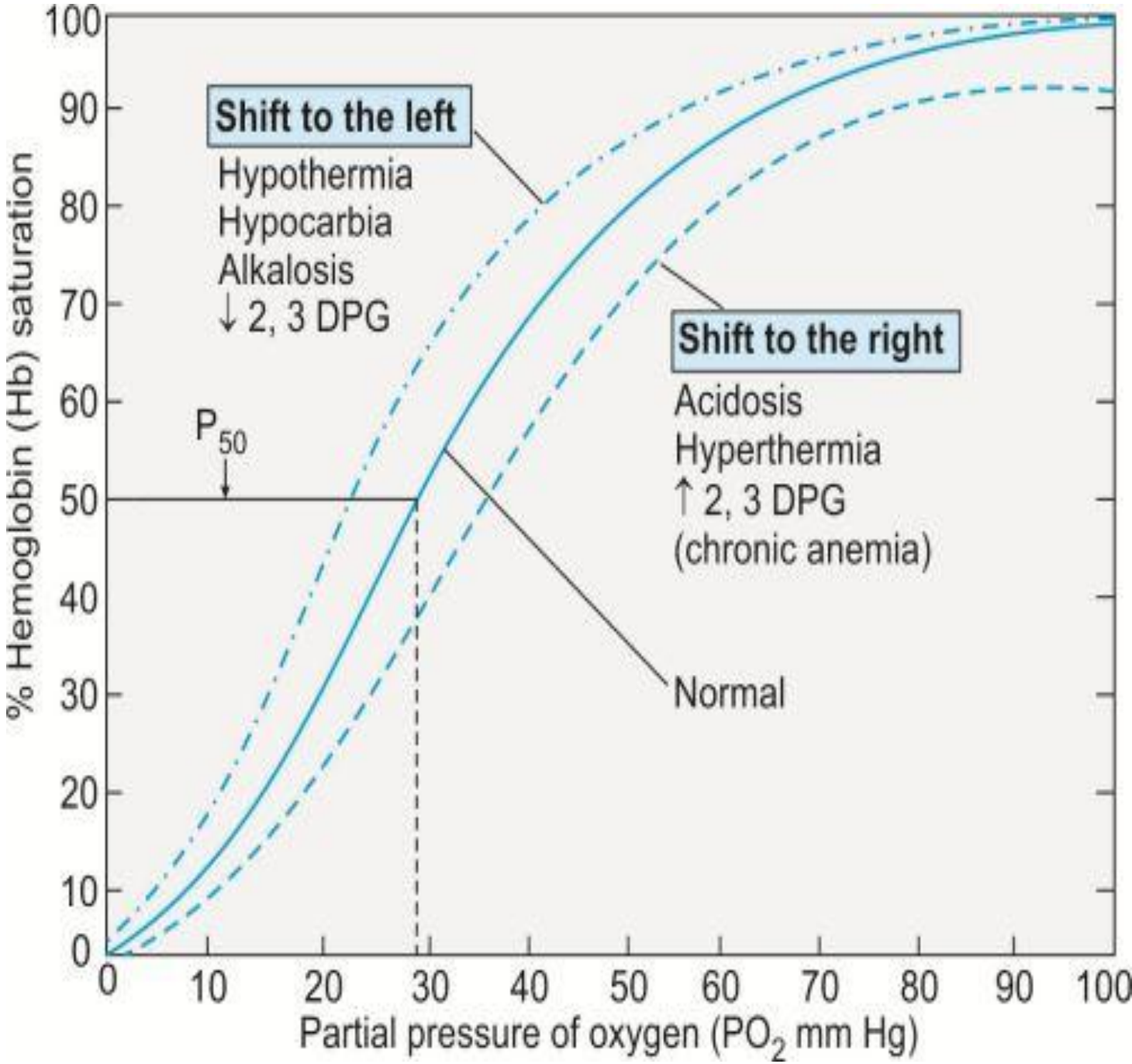
OXYGEN-HEMOGLOBIN DISSOCIATION CURVE

- It demonstrates the relationship between partial pressure of oxygen & percentage saturation of hemoglobin with oxygen (hemoglobin's affinity for oxygen).
- Normally, hemoglobin in the blood is saturated with oxygen upto 95%.
- Saturation of hemoglobin with oxygen depends upon the partial pressure i.e. when PO_2 is more, hemoglobin accepts oxygen & when the PO_2 falls, hemoglobin releases oxygen.
- Under normal conditions, oxygen-hemoglobin dissociation curve is *'S'* shaped (sigmoid shaped).
- Lower part of curve demonstrates dissociation of oxygen from hemoglobin whereas upper part of the curve indicates the uptake of oxygen by hemoglobin as the partial pressure of oxygen rises.

What is P_{50}?

It is the partial pressure of oxygen at which saturation of hemoglobin with oxygen is 50%.

- Hemoglobin is about 50% saturated when PO_2 is ***27 mm Hg***.
- At 40 mm Hg, the saturation is 75% whereas at 100 mm Hg, the saturation becomes 95%.



FACTORS AFFECTING OXYGEN-HEMOGLOBIN DISSOCIATION CURVE

- Oxygen-hemoglobin dissociation curve is shifted to left or right by various factors.
- ***Rightward shift*** indicates ***dissociation of oxygen*** from hemoglobin.
- ***Leftward shift*** demonstrates ***acceptance of oxygen*** by hemoglobin.

RIGHTWARD SHIFT

- Rightward shift of oxygen-hemoglobin dissociation curve shows ***lower affinity*** of hemoglobin for oxygen making it easier to unload oxygen for the active tissues.
- Oxygen-hemoglobin dissociation curve is shifted to right in conditions like:

i. Decrease in PO_2
ii. Increase in PCO_2
iii. Increase in H^+ ion concentration & decrease in pH (acidity)
iv. Increased body temperature
v. Excess of 2,3-BPG (byproduct in Embden-Meyerhof pathway of carbohydrate metabolism)

LEFTWARD SHIFT

- Shift of oxygen-hemoglobin dissociation to the left side indicates hemoglobin holds oxygen more tightly (***stronger affinity***) making it harder to release oxygen for the tissues.
- Oxygen-hemoglobin dissociation curve is shifted to left in conditions like:

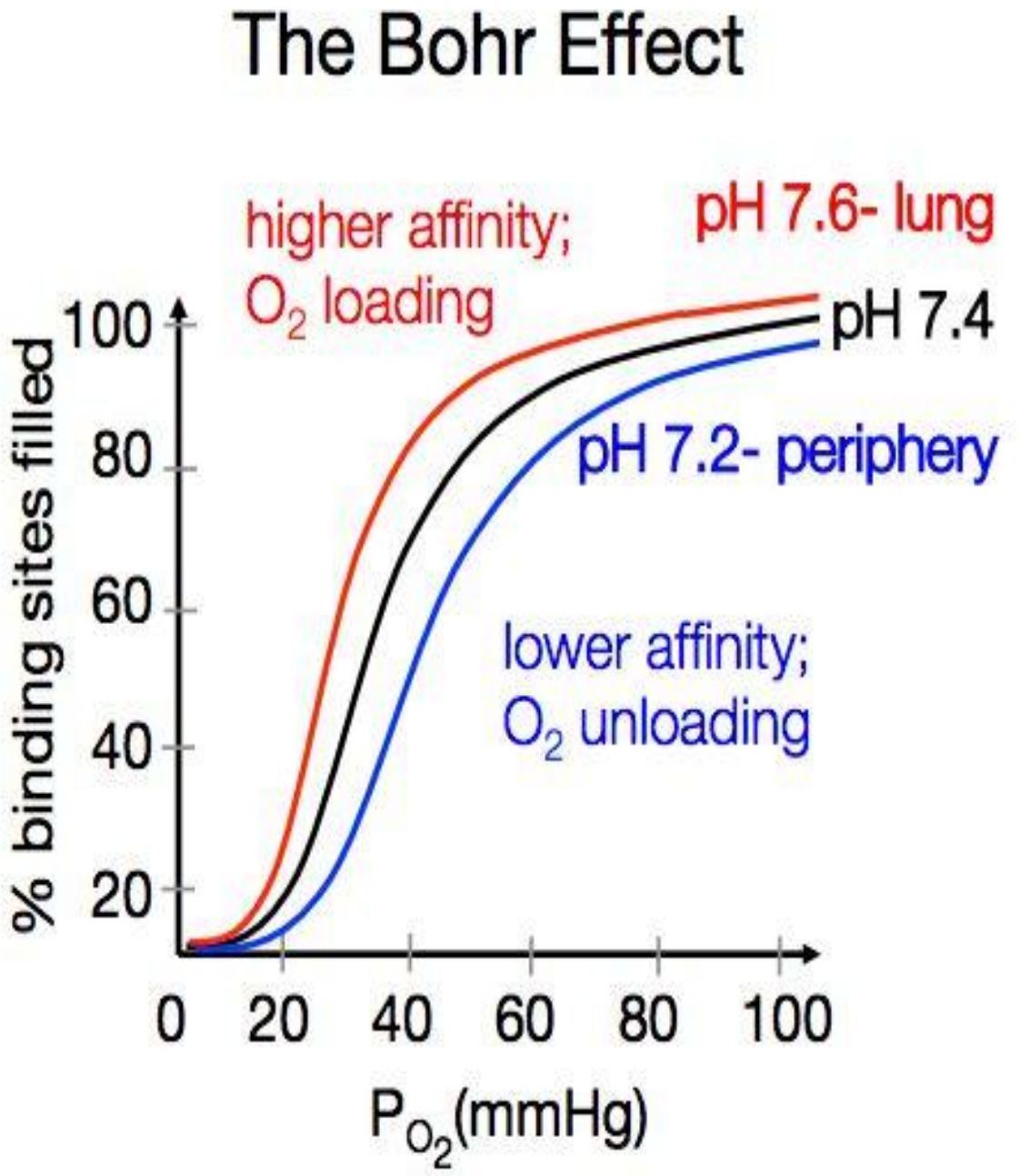
i. Increased pH (alkalinity)
ii. Decreased PCO_2
iii. Decreased temperature (hypothermia)
iv. Decreased 2,3-BPG
v. Combination of hemoglobin with CO (carbon monoxide)
vi. Fetal hemoglobin (HbF)

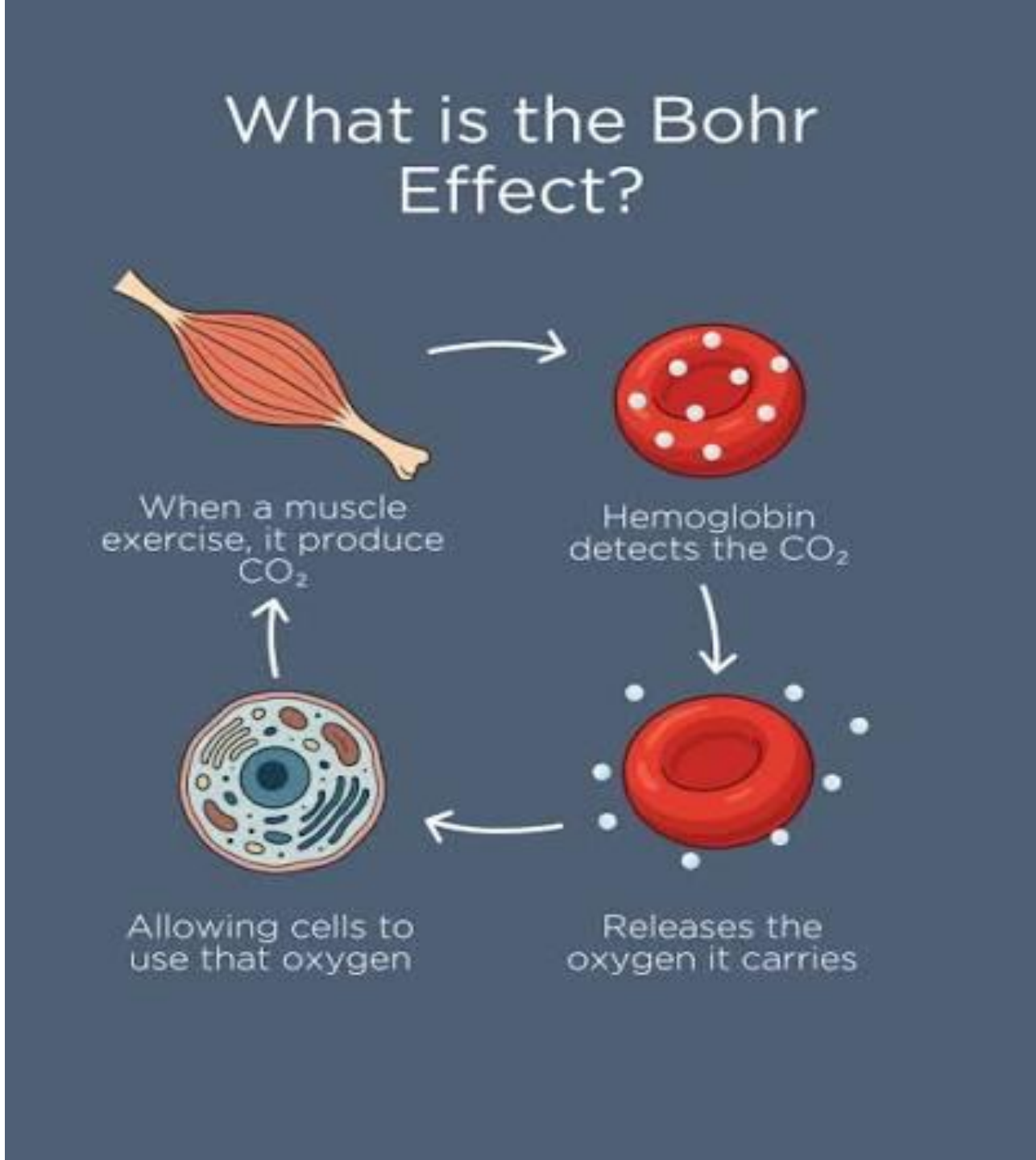
BOHR EFFECT

- Bohr effect shows that:

Presence of carbon dioxide (CO_2) decreases the affinity of hemoglobin for oxygen.

- In the tissues, due to continuous metabolic activities, partial pressure of carbon dioxide is very high whereas the partial pressure of oxygen is low.
- This pressure gradient causes CO_2 to enter the blood & O_2 is released from the blood to the tissues because CO_2 decreases the affinity of hemoglobin for O_2.
- As Bohr effect enhances the release of oxygen to the tissues, it shifts the oxygen-hemoglobin dissociation curve to the right.


The Bohr Effect
higher affinity;
O_2 loading
pH 7.6- lung
pH 7.4
pH 7.2- periphery
lower affinity;
O_2 unloading
% binding sites filled
100
80
60
40
20
0
20
40
60
80
100
P_{O_2}(mmHg)

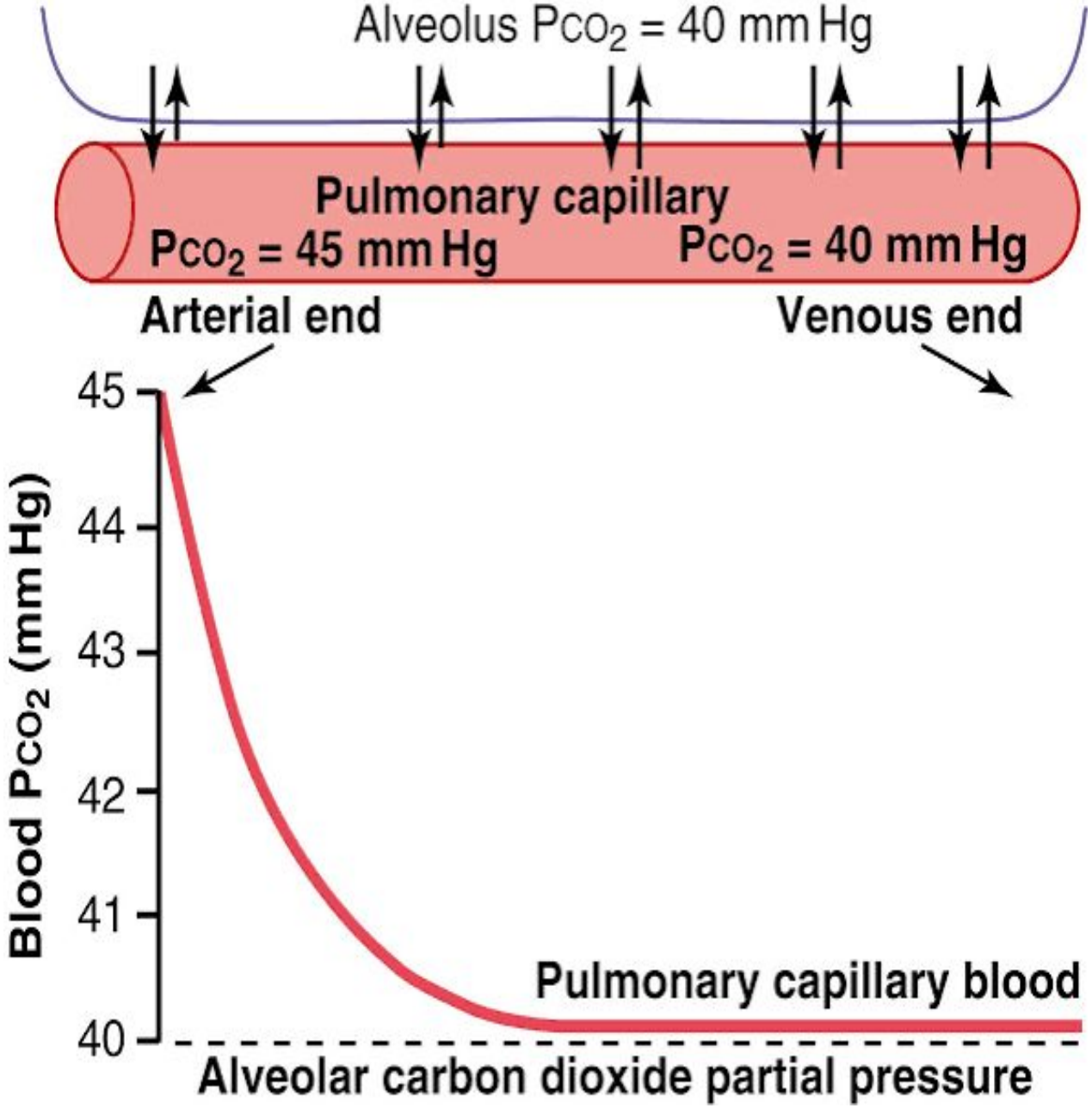

What is the Bohr Effect?
When a muscle exercise, it produce CO_2
Hemoglobin detects the CO_2
Releases the oxygen it carries
Allowing cells to use that oxygen

DIFFUSION OF CO_2 FROM TISSUE CELLS INTO CAPILLARIES

- When oxygen is used by the cells, virtually all of it is converted into carbon dioxide which increases the intracellular PCO_2.
- This elevation in PCO_2 causes carbon dioxide to diffuse from cells into the capillaries & then it is transported by the blood to the lungs where it diffuses from capillaries into the alveoli & finally expelled out(expiration).
- At each point in the gas transport chain, carbon dioxide diffuses exactly in the opposite direction as diffusion of oxygen.
- However, the major difference between O_2 & CO_2 is that CO_2 can diffuse about 20 times as rapidly as O_2.
- Therefore, the pressure gradient required for the diffusion of CO_2 is far less than the pressure differences required to cause diffusion of O_2.

Cont..

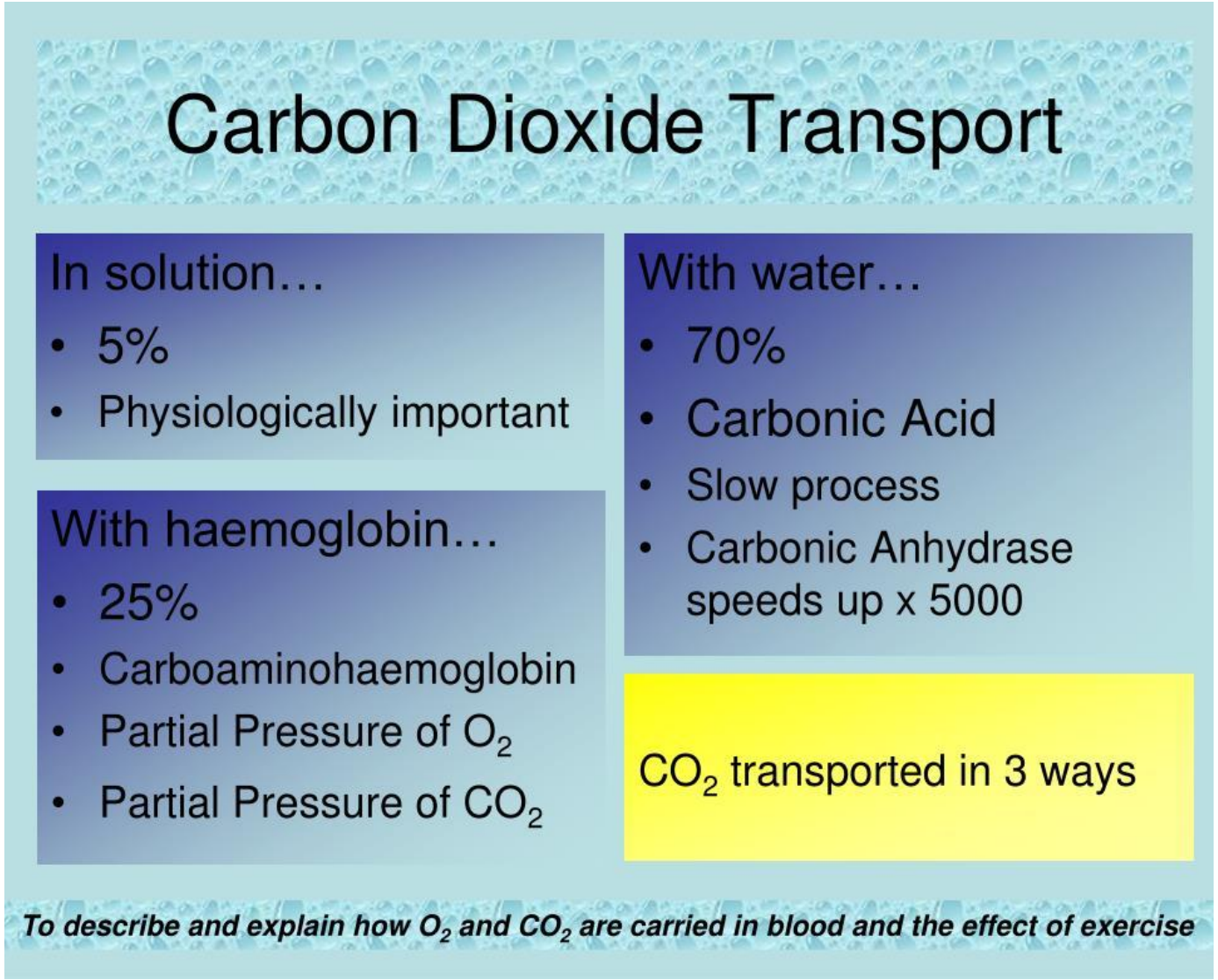
- Intracellular PCO_2 is 46 mm Hg whereas interstitial PCO_2 is 45 mm Hg, so the pressure difference is only 1 mm Hg.
- PCO_2 of arterial blood entering the tissues is 40 mm Hg & PCO_2 of venous blood leaving the tissues is 45 mm Hg so PCO_2 of blood entering the pulmonary capillaries is 45 mm Hg & PCO_2 of alveolar air is 40 mm Hg.
- Thus, pressure gradient of only 5 mm Hg is sufficient to cause all the required diffusion of carbon dioxide out of pulmonary capillaries into the alveoli.



TRANSPORT OF CARBON DIOXIDE (CO_2)

- CO_2 is transported by the blood from cells to the lungs.
- Carbon dioxide is transported in the blood in four ways i.e.

1) As dissolved form (7%)
2) As carbonic acid (negligible)
3) As bicarbonate (63%)
4) As carbamino compounds (30%)



TRANSPORT OF CO_2 IN DISSOLVED FORM

- CO_2 diffuses into the blood & dissolves in the fluid of plasma forming a simple solution.
- However, only about 3 ml/100 ml of plasma, carbon dioxide is transported in this way which constitutes only about 7% of total CO_2 in the blood.

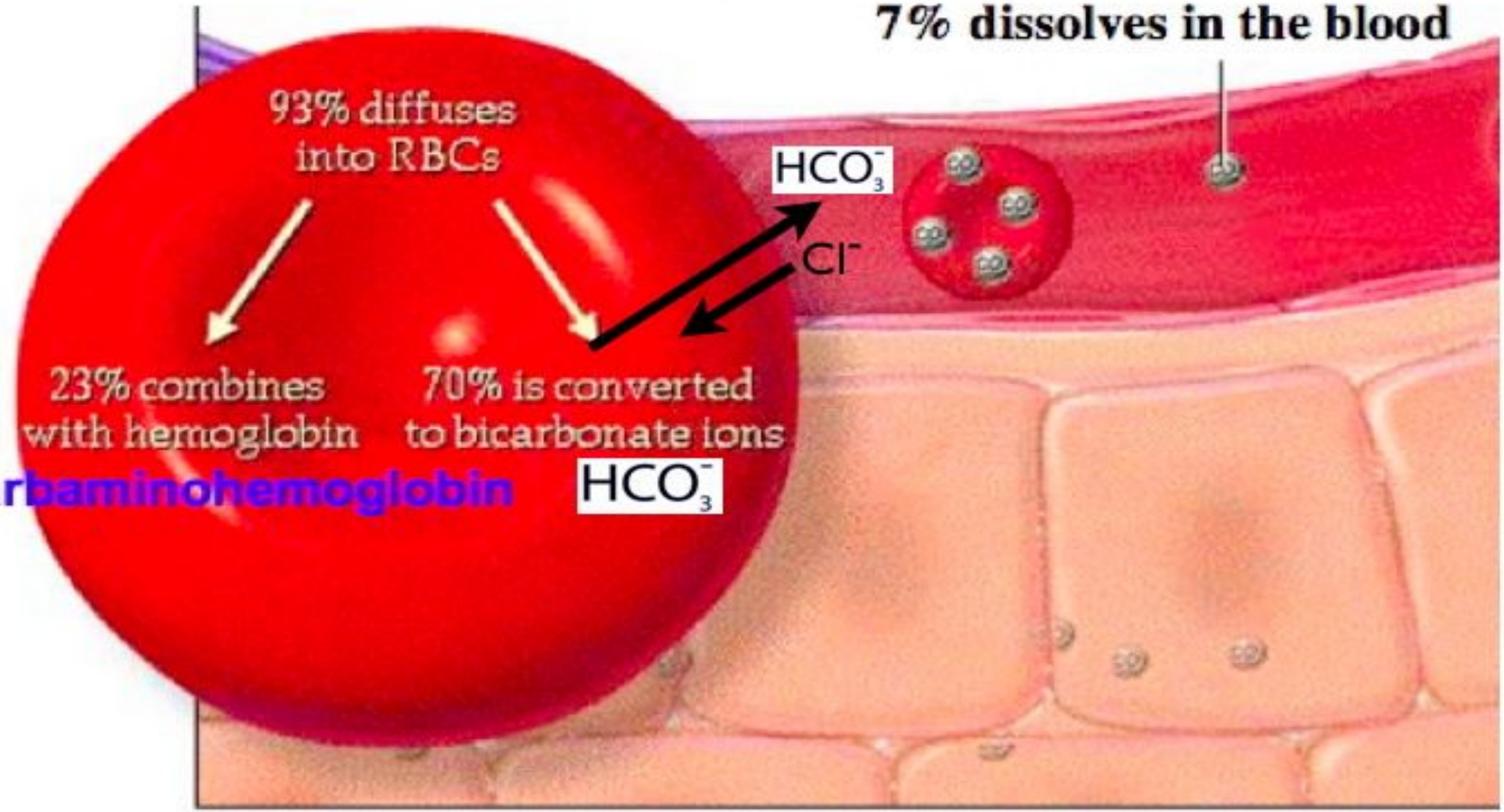
TRANSPORT OF CO_2 AS CARBONIC ACID

- Part of dissolved carbon dioxide in plasma combines with water to form carbonic acid.
- However, amount of carbon dioxide transported in this way is very negligible.

TRANSPORT OF CO_2 AS BICARBONATE

- About 63% of total CO_2 is transported in the form of bicarbonate.
- Carbon dioxide enters the RBCs from plasma & then inside the RBCs CO_2 combines with water to form carbonic acid.
- The reaction inside the RBCs is very quick due to the presence of carbonic anhydrase which speeds up the reaction.
- As carbonic anhydrase is present only in RBCs & not in plasma, formation of carbonic acid is atleast 200 to 300 times more in RBCs than in plasma.
- Because of unstable nature of carbonic acid, almost all carbonic acid formed in the RBCs dissociates into bicarbonate & hydrogen ions.
- As the concentration of bicarbonate ions increases inside the RBCs, they diffuse through the cell membrane into the plasma.

CO_2 Diffusing from Tissues



CHLORIDE SHIFT (HAMBURGER PHENOMENON)

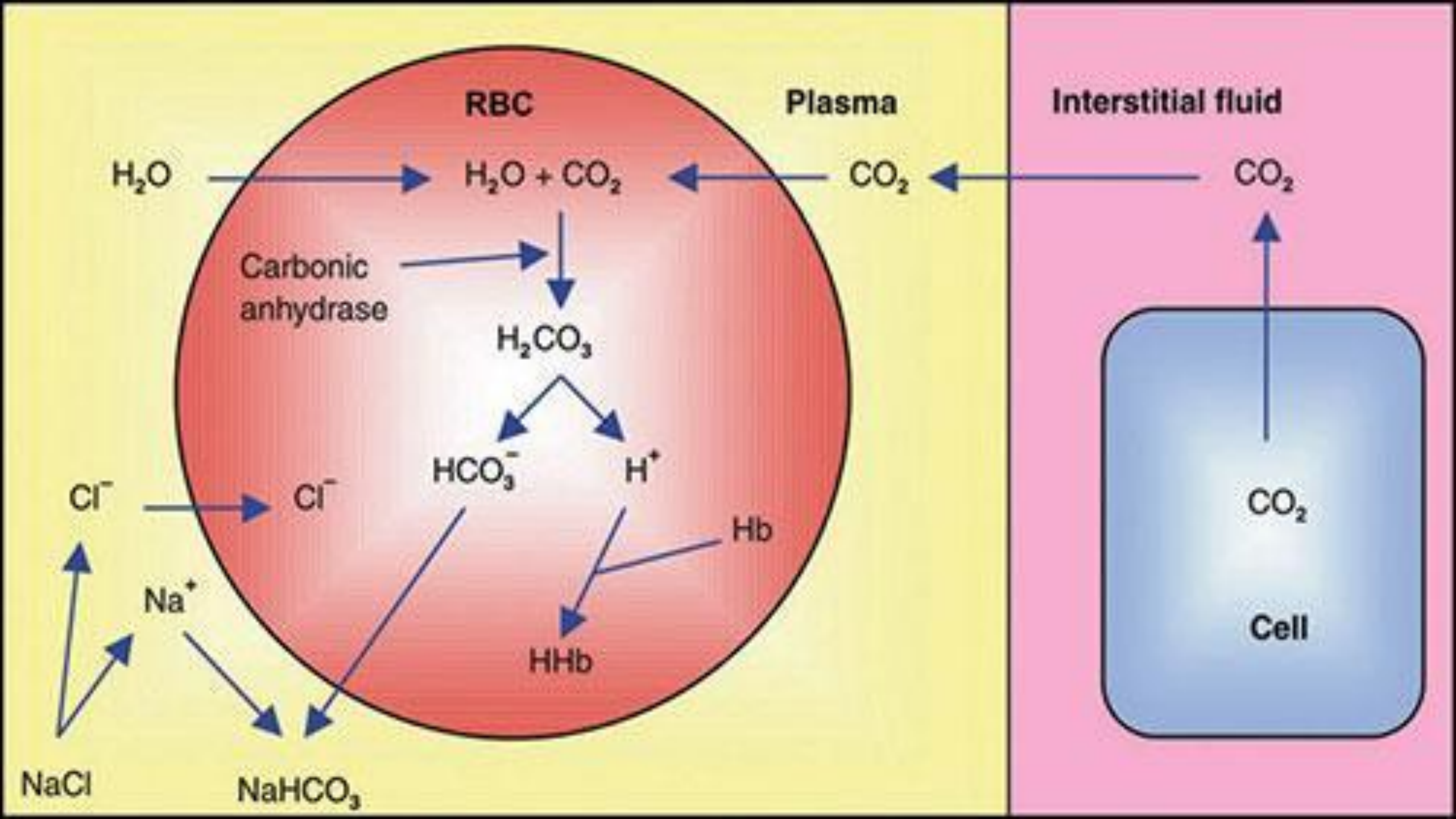
It is the exchange of chloride ion for a bicarbonate ion across the membrane of RBCs.

- Chloride shift occurs when CO_2 enters the blood from the tissues.
- In plasma, large quantity of sodium chloride(NaCl) is present which dissociates into sodium(Na^+) & chloride ions(Cl^-).
- When the negatively charged bicarbonate ions(HCO_3^-) move out of RBCs into the plasma, the negatively charged chloride ions(Cl^-) move into the RBCs to maintain ionic balance (electrolyte equilibrium).
- ***Band 3 protein*** also called ***anion exchanger 1 (AE1)*** acts like a pump in RBC membrane & is responsible for the exchange of bicarbonate & chloride ions.
- Bicarbonate ions(HCO_3^-) combine with Na^+ ions in the plasma to form ***sodium bicarbonate*** & transported in the blood.
- H^+ ions formed from carbonic acid are buffered by hemoglobin inside the RBCs.

REVERSE CHLORIDE SHIFT

The process by which chloride ions (Cl-) are moved back into the plasma from RBC shift.

- It occurs in the lungs & helps in the elimination of CO_2 from the blood.
- When blood reaches the alveoli of lungs, sodium bicarbonate in the plasma of blood dissociates into Na^+ & HCO_3^-.
- Bicarbonate ions move into the RBC & chloride ions move out of the RBC into the plasma.
- In the plasma, chloride ions(Cl^-) combines with sodium(Na^+) to form sodium chloride(NaCl) again.
- Bicarbonate ion inside the RBC combines with hydrogen ion to form carbonic acid which ultimately dissociates into water & carbon dioxide.
- Carbon dioxide is then expelled out into the air.

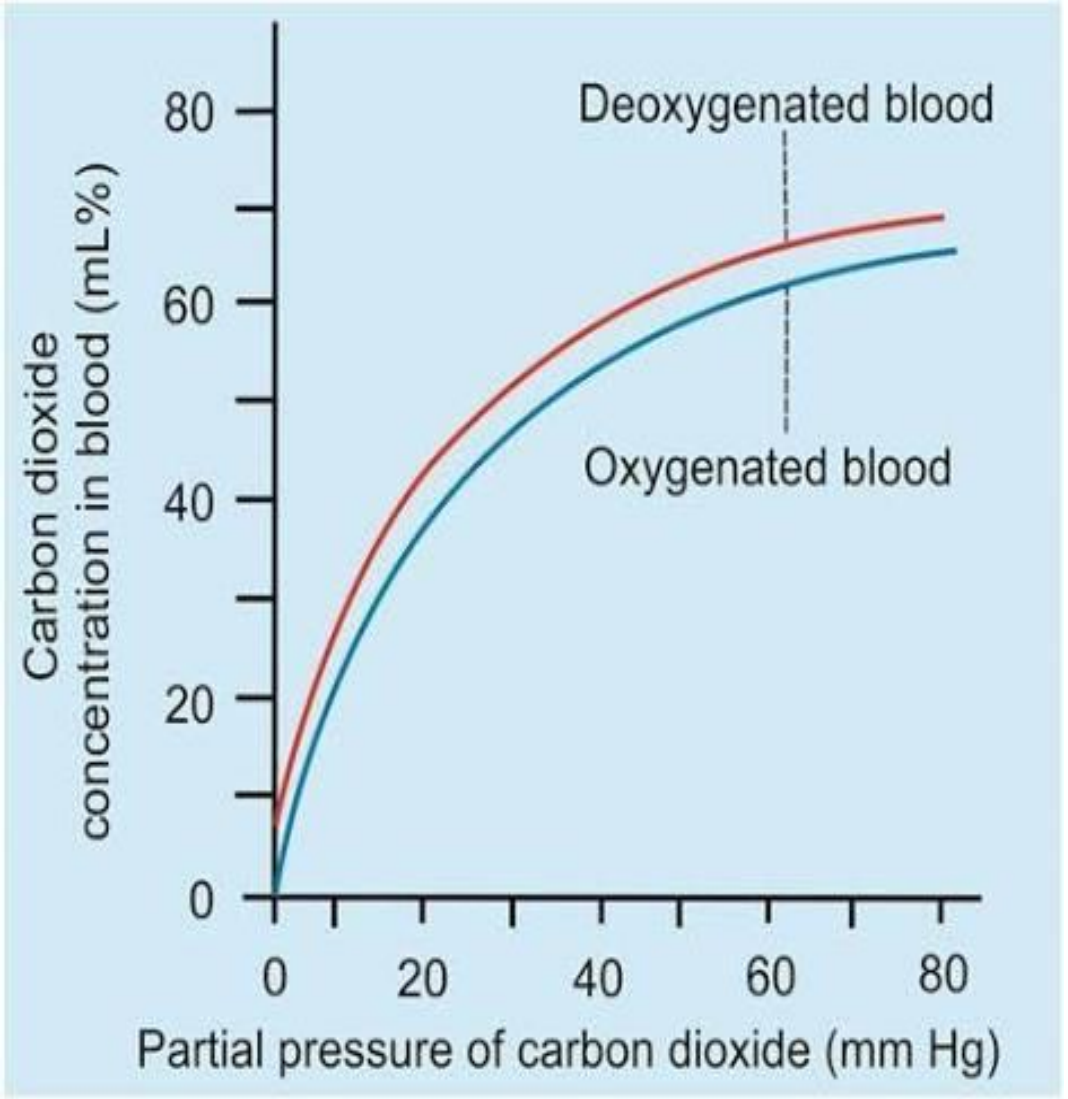

RBC
Plasma
Interstitial fluid
H_2O
$H_2O + CO_2$
CO_2
CO_2
Carbonic anhydrase
H_2CO_3
HCO_3^-
H^+
Cl^-
Cl^-
Hb
Na^+
HHb
CO_2
Cell
NaCl
$NaHCO_3$

TRANSPORT OF CO_2 AS CARBAMINO COMPOUNDS

- About 30% of total carbon dioxide is transported in the form of carbamino compounds i.e. CO_2 is transported in the blood in combination with hemoglobin & plasma proteins.
- Carbon dioxide combines with hemoglobin to form ***carbamino- hemoglobin*** (***carbhemoglobin***) & with plasma proteins to form ***carbamino proteins***.
- Carbamino hemoglobin & carbamino proteins are collectively termed as ***carbamino compounds***.
- Carbon dioxide combines with plasma proteins & hemoglobin with a loose band so that CO_2 can be easily released into the alveoli due to partial pressure gradient.
- Amount of carbon dioxide transported in combination with plasma proteins is very less as compared to the amount that is transported in combination with hemoglobin.
- It is because, the quantity of proteins in plasma is only half the quantity of hemoglobin.

CARBON DIOXIDE DISSOCIATION CURVE

- Carbon dioxide is transported in the blood as physical solution & in combination with water, plasma proteins & hemoglobin.
- The amount of carbon dioxide combining with blood depends upon the partial pressure of carbon dioxide.
- Carbon dioxide dissociation curve shows the relationship between partial pressure of CO_2 & the amount of CO_2 that combines with the blood.
- This curve indicates that carbon dioxide content in blood is 48 ml/100 ml of blood when PCO_2 is 40 mm Hg & 52 ml/100 ml of blood when the PCO_2 is 48 mm Hg.
- When PCO_2 is about 100 mm Hg, carbon dioxide content in the blood is 70 ml/100 ml of blood.



HALDANE EFFECT

- Haldane effect shows that:

Combination of oxygen with hemoglobin displaces carbon dioxide from hemoglobin i.e. excess of oxygen content in the blood causes a rightward shift of carbondioxide dissociation curve.

- When hemoglobin combines with oxygen, it becomes strongly acidic which causes displacement of carbon dioxide from hemoglobin(acidic Hb has low tendency for CO_2).
- Hydrogen ions are released in excess due to acidity which combines with bicarbonate ions to form carbonic acid.
- Carbonic acid in turn dissociates into water & carbon dioxide which is finally released from blood into the alveoli.

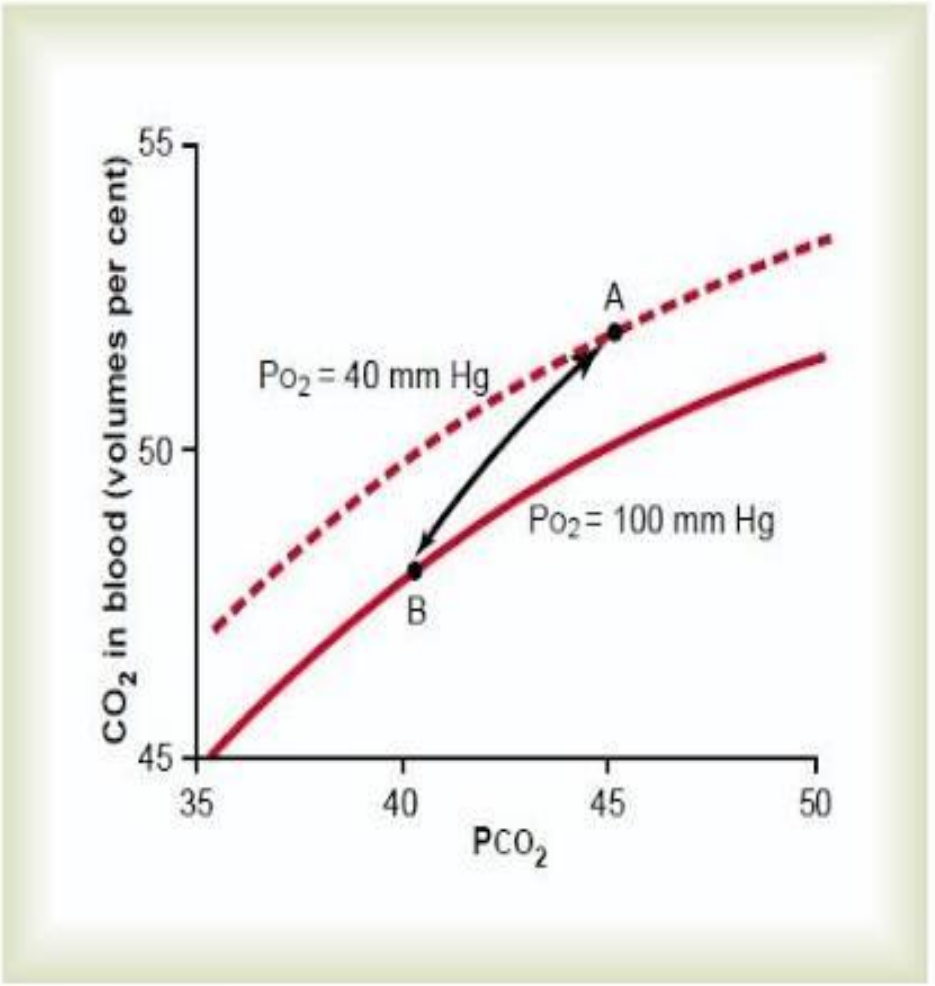


Figure 40–15

Portions of the carbon dioxide dissociation curve when the P_{O_2} is 100 mm Hg or 40 mm Hg. The arrow represents the Haldane effect on the transport of carbon dioxide, as discussed in the text.

POST ASSESSMENT

Q1. The oxygen carrying capacity of 1 gram of hemoglobin is approximately:

a) 0.34 ml
b) 3.4 ml
c) 1.34 ml
d) 13.4 ml
e) 20 ml

Q2. In systemic tissue capillaries, bicarbonate ions leave RBCs in exchange for:

a) Sodium ions
b) Potassium ions
c) Hydrogen ions
d) Chloride ions
e) Calcium ions

Cont..

Q3. During exercise, active skeletal muscles produce more CO2 & heat. What is the principal effect on oxygen unloading from hemoglobin?

a) A leftward shift & reduced oxygen unloading
b) A rightward shift & increased oxygen unloading
c) Increased hemoglobin affinity due to increased temperature
d) Complete saturation of hemoglobin in tissues
e) No change in oxygen affinity

Q4. Approximately what percentage of oxygen is transported bound to hemoglobin under normal conditions?

a) 10%
b) 25%
c) 50%
d) 75%
e) 98%

Thank You

Any Questions?